Program Evaluation with Observational Data

Chapter 5 - Difference-in-Difference

Advanced Empirical Research Method

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Introduction

General Context:

Treatment introduced at one point in time.
Outcomes and covariates observed **before** and **after** the treatment.

To make the before/after comparison ‘causal’, it is important to have a **control group**, unaffected by the treatment.

Difference-in-Difference estimators are very popular for the evaluation of regional policies.
Key assumption: Without treatment, the trend of treated and untreated region would have remained the same.

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Basic set-up

2 regions: $R \in \{0,1\}$ of which one ($R = 1$) is exposed to treatment.
2 time periods, $t \in \{0,1\}$ where $t = 0$ is before treatment, $t = 1$ after treatment.

Sampling: There are many 'units' in each region R.

Assumption: No composition effect between R (e.g. no in or out migration).

The treatment variable is:

$$D = Rt.$$

The potential outcomes are $Y_t(d)$ for $d \in \{0,1\}$ and $t \in \{0,1\}$.
The observed outcome is:

$$Y_t = (1 - D)\,Y_t(0) + DY_t(1)$$

Average Treatment on the Treated

$$ATT = E(Y_1(1) - Y_1(0)|R = 1)$$

$$\begin{aligned}
ATT &= E(Y_1(1) - Y_0(0)|R = 1) - E(Y_1(0) - Y_0(0)|R = 1) \\
&= E(Y_1(1) - Y_0(0)|R = 1) - E(Y_1(0) - Y_0(0)|R = 0) \\
&\quad + E(Y_1(0) - Y_0(0)|R = 0) - E(Y_1(0) - Y_0(0)|R = 1) \\
&= E(Y_1 - Y_0|R = 1) - E(Y_1 - Y_0|R = 0) \\
&\quad + E(Y_1(0) - Y_0(0)|R = 0) - E(Y_1(0) - Y_0(0)|R = 1)
\end{aligned}$$

Average Treatment on the Treated

A desirable assumption is:

$$E\left(Y_1(0) - Y_0(0) | R = 0\right) = E\left(Y_1(0) - Y_0(0) | R = 1\right)$$

Equal trend assumption:
In the absence of treatment, the evolution of potential outcomes would have been the same in both regions.

This assumption is (partly) testable if one observes more than one period before treatment.

Average-Treatment on the Treated

An alternative interpretation:

$$E(Y_1(0)|R=1) - E(Y_1(0)|R=0) = E(Y_0(0)|R=1) - E(Y_0(0)|R=0)$$

The relative differences between regions is kept constant over time. DiD allows regional unobserved confounders, which are constant over time.

For comparison, **panel data** can handle individual unobserved confounders, which are constant over time.

Average Treatment on the Treated

An additional, often implicit assumption is the 'no anticipation' assumption:

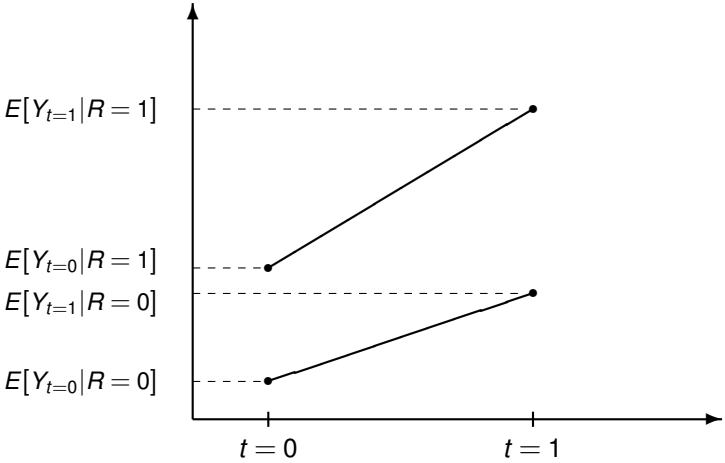
$$E\left(Y_0(1)|R=1\right)=E\left(Y_0(0)|R=1\right)$$

Under the 'equal trend' assumption and the 'no anticipation' assumption:

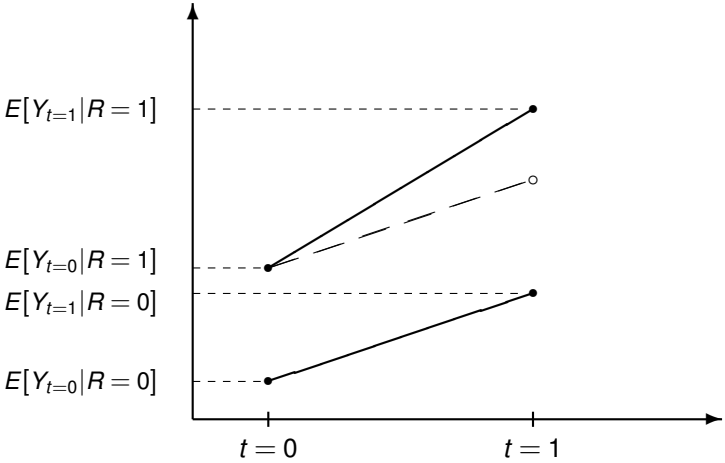
$$ATT=E\left(Y_1-Y_0|R=1\right)-E\left(Y_1-Y_0|R=0\right)$$

which is called the difference-in-difference or diff-in-diff quantity.

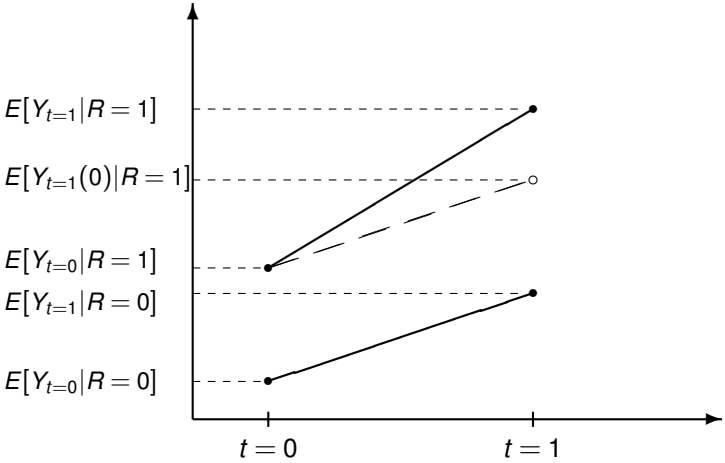
Difference-in-Differences: Graphical Interpretation



Difference-in-Differences: Graphical Interpretation



Difference-in-Differences: Graphical Interpretation



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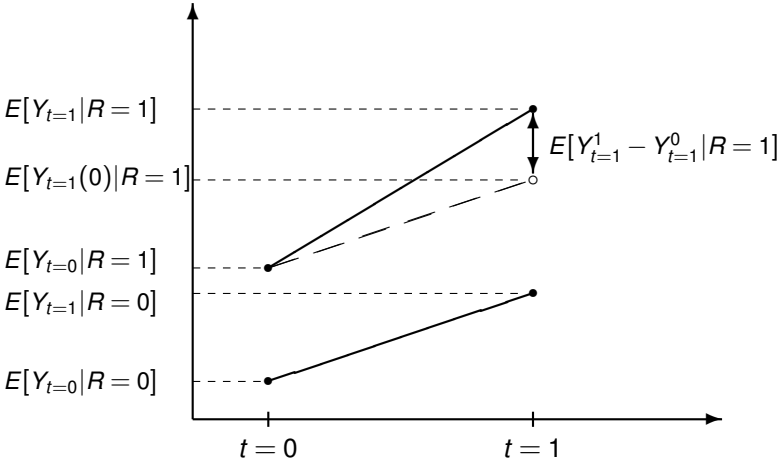

Identification in a linear-model

It is instructive to review the condition for identification in a linear model.

It is always possibles to write the potential outcome as:

$$Y_t(d) = \alpha + \beta \cdot d + \gamma R + \delta t + \theta Rt + U_t(d)$$

where $E(U_t(d)|R) = 0$ for $d = 0, 1$ and $t = 0, 1$.

With these notations, the equal trend assumption writes:

$$E\left(Y_1(0) - Y_0(0)|R = 1\right) = \delta + \theta = \delta = E\left(Y_1(0) - Y_0(0)|R = 1\right)$$

which implies that $\theta = 0$. There is no interaction effect between the cross section dimension R and the time dimension t.

Estimation in a linear-model

A regression model can be written by observing that :

$$\begin{aligned} Y_t &= Y_t(0) + D(Y_t(1) - Y_t(0)), \\ &= \alpha + \gamma R + \delta t + U_t(0) + D(\beta + U_t(1) - U_t(0)) \\ &= \alpha + \gamma R + \delta t + D\beta + \underbrace{(U_t(0) + D(U_t(1) - U_t(0)))}_{:=V_t}, \text{ where } E(V_t|R, D) = 0. \end{aligned} \tag{1}$$

Hence, the OLS estimate for:

$$Y_t = \alpha + \gamma R + \delta t + D\beta + V_t$$

delivers a consistent estimate of $ATT = E(Y_1(1) - Y_1(0)|R = 1) = \beta$.

Example: Minimum wage and employment

Card and Krueger (1994) estimated the impact of an increase in the minimum wage

- In April 1, 1992 the minimum wage increased from $4.25 to $5.05 in New Jersey;
- In Pennsylvania (the neighboring state) the minimum wage remained at $4.25.

The authors collected employment data at 410 fast food restaurants around the border.
For replication:
`https://rpubs.com/phle/r_tutorial_difference_in_differences`

Example: Minimum wage and employment

Potential outcomes:

$Y_t(1)$ = fast food employment at a restaurant in a state if there is a high state minimum wage (New Jersey).
$Y_t(0)$ = fast food employment at a restaurant in a state if there is a low state minimum wage (Pennsylvania).

Observed outcomes:

$$Y_t = \alpha + \gamma NJ + \delta Nov + \beta(NJ \cdot Nov) + V_t$$

DiD parameter:

$$\begin{aligned}\beta = \quad & [E(Y_t|NJ=1, Nov=1) - E(Y_t|NJ=1, Nov=0) \\ & - [E(Y_t|NJ=0, Nov=1) - E(Y_t|NJ=0, Nov=0)]\end{aligned}$$

Example: Minimum wage and employment

Variable	PA (i)	NJ (ii)	Difference, NJ-PA (iii)
1. FTE employment before, all available observations	23.33 (1.35)	20.44 (0.51)	-2.89 (1.44)
2. FTE employment after, all available observations	21.17 (0.94)	21.03 (0.52)	-0.14 (1.07)
3. Change in mean FTE employment	-2.16 (1.25)	0.59 (0.54)	2.76 (1.36)

Notes: Adapted from Card and Krueger (1994), Table 3. The table reports average full-time equivalent (FTE) employment at restaurants in Pennsylvania and New Jersey before and after a minimum wage increase in New Jersey. The sample consists of all stores with data on employment. Employment at six closed

Example: Minimum wage and employment

Employment in Pennsylvania restaurants is somewhat higher than in New Jersey in February but falls by November. Employment in New Jersey, in contrast, increases slightly.

These two changes produce a positive difference-in-difference effect, the opposite of what we might expect if a higher minimum wage pushes businesses up the labor demand curve.

The key identifying assumption here is that employment trends would be the same in both states in the absence of treatment, which is untestable with only two periods (pre and post treatment).

The treatment induces a deviation from this common trend.

A few examples

Levine, Toffel, and Johnson (2012)

Research question: Does the enforcement of workplace safety regulations improve workplace safety?

Compares randomly inspected establishments in California (treatment group) with similar establishments, eligible but not chosen (control group).

	Treatment	Control	$\Delta\Delta$
Injuries before	3.69	3.05	
Injuries after	3.14	3.04	
Δ	-0.55	-0.01	-0.54

A few examples

Olson (2009)

Research question: Does Employment Protection Law (EPL) encourage absenteeism?

A Swedish 2001 reform increased the risk of redundancy, especially for workers with high amount of sickness absence. Compare firms with 2-9 workers (treatment) to firms with 12-50 whose EPL was unaffected.

	Treatment	Control	$\Delta\Delta$
Prob. of absence before	2.8	3.6	
Prob. of absence after	2.4	3.6	
Δ	-0.4	0.0	-0.4

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Extensions of the basic set up

There are several avenues to go beyond the 2 by 2 case:

1. Allow for additional covariates.
2. Allow for multiple time periods (i.e. $T > 2$);
3. Allow for multiple regions (i.e. $R > 2$);
4. Allow for heterogeneous treatment times;

Extension: Covariates

Assume that there are (regional and time-varying) confounders X for the trends. We need:

Equal trend:	$E(Y_1(0) - Y_0(0)\|X, R = 1) = E(Y_1(0) - Y_0(0)\|X, R = 0)$
No bad control:	$X(1) = X(0) = X$
No anticipation:	$E(Y_0(1)\|X = x, R = 1) = E(Y_0(0)\|X = x, R = 1)$
Overlap:	$\Pr(D = 1, T = 1\|X, (D, t) \in \{(d, t), (1, 1)\}) < 1$ for all (d, t)

Under this modified equal trend assumption:

$$ATT(x) = E(Y_1 - Y_0|X = x, R = 1) - E(Y_1 - Y_0|X = x, R = 0)$$

which is the conditional treatment on the treated parameter.

Extension: Covariates

It is worth noting that we are back in the framework of 'selection-on-observables'. Only now, the 'outcome' variable is the trend $(Y_1(d) - Y_0(d))$ not the level.

Therefore, the previous estimation strategies apply:

1. A regression model for the outcome variable $\mu_d(t,x) = E(Y_t|D=d, X=x)$:

$$ATT = E\left(\mu_1(1,X) - \mu_1(0,X) - (\mu_0(1,X) - \mu_0(0,X))|D=1, T=1\right)$$

2. Inverse Probability weighting:

$$ATT = \quad E\left[\left\{\frac{D.T}{p_{11}} - \frac{D.(1-T)p_{11x}}{p_{10x}p_{11}} - \frac{(1-D).T}{p_{01x}p_{11}} - \frac{(1-D).(1-T)p_{11x}}{p_{00x}p_{11}}\right\}Y\right]$$

with $p_{11} = \Pr(D=1, T=1)$ and $p_{dtx} = \Pr(D=d, T=t|X)$

Extension: Covariates

It is worth noting that we are back in the framework of 'selection-on-observables'. Only now, the 'outcome' variable is the trend $(Y_1(d) - Y_0(d))$ not the level.

Therefore, the previous estimation strategies apply:

3. A doubly-robust estimator; see Zimmert (2020);
4. Double/Debiased Machine learning to control for covariates in a data driven way.
5. Linear-model with linearly included covariates: this is the most common and usually default solution. This imposes strong assumptions about treatment effect homogeneity.

Extension: Multiple time periods

Pre-treatment period: from $t = 1$ to $t_d - 1$; Treatment period: from t_d until T.

As previously and without loss of generality, it is possible to write the potential outcome:

$$Y_t(d) = \alpha + \delta_t + \beta_t \cdot d + \gamma R + \sum_{t^*=1}^{T} \theta_{t^*} R I\{t^* = t\} + U_t(d)$$

where $E(U_t(d)|R) = 0$ for $d = 0, 1$ and $t = \in \{1, \cdots, T\}$, and the parameters (δ_t, θ_t) are time-specific parameters. By assumption: $\delta_1 = 0$

The assumption of equal trend boils down to:

$$\theta_t = 0, \forall t.$$

Extension: Multiple time periods

As expected, β_t for $t \leq t_d$ are identified.

The estimation by OLS proceeds as in the elementary case. That is, one runs the regression:

$$Y_t = \alpha + \delta_t + \gamma R + D_t \beta_t + V_t$$

The OLS estimators of β_t are consistent.
It is common furthermore to assume that $\beta_t = \beta$ and this can be tested.

Extension: A placebo test

The adequate regression model to estimate is:

$$Y_t = \alpha + \delta_t + \gamma R + D_t \beta_t + V_t$$

Suppose now that we choose $t^* = t \in \{1, \cdots, t_{d-1}\}$ and write $D_t^* = RI\{t = t^*\}$.

In the regression

$$Y_t = \alpha + \delta_t + \gamma R + D_t \beta + D_t^* \beta^* + V_t$$

We should have $\beta^* = 0$ since t^* belongs to a pre-treatment period.

Example: Minimum Wage (continued)

In an update of their original minimum wage study, Card and Krueger (2000) obtained administrative payroll data for restaurants in New Jersey and Pennsylvania for a number of years.

In October 1996, the federal minimum wage increased from $4.25 to $4.75 in Pennsylvania and remained unchanged (at $5.05) in New Jersey.

These data give us an opportunity to look at a new minimum wage“experiment”.

Example: Minimum Wage (continued)

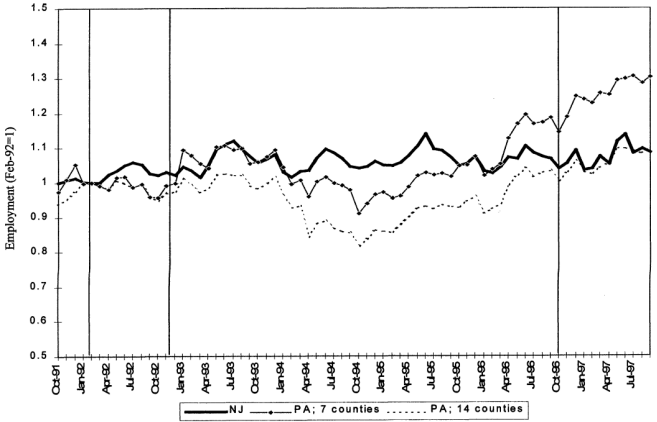


FIGURE 2. EMPLOYMENT IN NEW JERSEY AND PENNSYLVANIA FAST-FOOD RESTAURANTS, OCTOBER 1991 TO SEPTEMBER 1997

Note: Vertical lines indicate dates of original Card-Krueger survey and the October 1996 federal minimum-wage increase.
Source: Authors' calculations based on BLS ES-202 data.

Example: Minimum Wage (continued)

The administrative data show a slight decline in employment from February to November 1992 in Pennsylvania, and little change in New Jersey over the same period.

However, the data also reveal fairly substantial year-to-year employment variation in other periods. These swings often seem to differ substantially in the two states.

So Pennsylvania may not provide a very good measure of counterfactual employment rates in New Jersey in the absence of a policy change, and vice versa.

Example: EPL

Bjuggren (2018) looks at the effects of 2001 Swedish Reform of EPL on firms' productivity: Employers received more flexibility to choose which worker to make redundant.

Compare firms with 2-9 workers (treatment) to firms with 12-50 whose EPL was unaffected.

Example: EPL

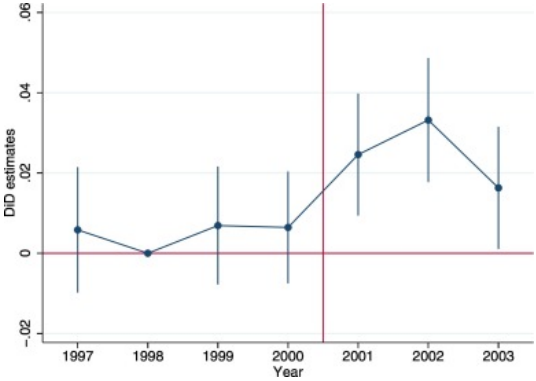


Figure: Effect of the 2001 EPL reform on firms' productivity

Extension: Multiple regions

Consider now that R is discrete in $\{1 \cdots \bar{R}\}$. We now have:

$$Y_t(d) = \alpha + \delta_t + \sum_{r=1}^{\bar{R}} \beta_{r,t} \cdot d + \sum_{r=1}^{\bar{R}} \gamma_r \mathbf{1}\{R = r\} + \sum_{r=1}^{\bar{R}} \sum_{t^*=1}^{T} \theta_{r,t^*} \mathbf{1}\{t^* = t\} \mathbf{1}\{R = r\} + U_t(d)$$

where $E(U_t(d)|R) = 0$ for $d = 0, 1$, $t = \in \{1, \cdots, T\}$, and $R \in \{1 \cdots \bar{R}\}$. The parameters (δ_t, θ_t) are time-specific parameters, and the parameters γ_r are region-specific. By assumption: $\delta_1 = \gamma_1 = 0$, and $\theta_{r,t} = 0, \forall r, t$ (parallel trend).

Extension: Multiple regions

$$Y_t(d) = \alpha + \delta_t + \sum_{r=1}^{\bar{R}} \beta_{r,t} \cdot d + \sum_{r=1}^{\bar{R}} \gamma_r \mathbf{1}\{R = r\} + \sum_{r=1}^{\bar{R}} \sum_{t^*=1}^{T} \theta_{r,t^*} \mathbf{1}\{t^* = t\} \mathbf{1}\{R = r\} + U_t(d)$$

Here Region 1 is the control region and all effect compared to Region 1. It is not really meaningful to compare treated regions between them because the treated populations are not the same. Besides, this representation does not fully 'exploit' the presence of multiple regions, as it estimates a different effect for each region.

A parsimonious representation uses $\beta_{r,t} = \beta_t, \forall r \geq 2$.

The estimation by OLS is consistent and proceeds as follows:

$$Y_t = \alpha + \delta_t + \sum_{r=1}^{\bar{R}} \gamma_r \mathbf{1}\{R = r\} + D_t \beta_t + V_t$$

Extension: Heterogeneous treatment times

The previous setup extends easily to the case in which the treatment times, say t_r , are different across regions so that in the model.

Define:

$$D_{rt} := \mathbf{1}\{R = r, t \geq t_r\}$$

The estimation by OLS proceeds as follows:

$$Y_t = \alpha + \delta_t + \sum_{r=1}^{\bar{R}} \gamma_r \mathbf{1}\{R = r\} + \sum_{\tau=\underline{\tau}}^{\overline{\tau}} D_{rt} \mathbf{1}\{t = t_r + \tau\} \beta_\tau + V_t$$

The OLS estimators of β_τ are consistent and represent the treatment effect after τ periods of treatment.

Cengiz et al. (2019)

Aim of the paper:

- **138 state-level** minimum wage changes **between 1979 and 2016** in the United States.
- Estimate the effect of the minimum wage increase on employment changes **by wage bins**, relative to the new minimum wage.
- Compare the number of excess jobs paying at or slightly above the new minimum wage to the missing jobs paying below it to infer the employment effect.
- The overall number of low-wage jobs remained essentially unchanged over the five years following the increase.

Cengiz et al. (2019)

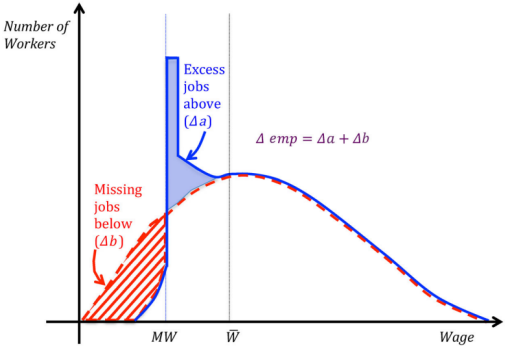


FIGURE I

The Impact of Minimum Wages on the Frequency Distribution of Wages

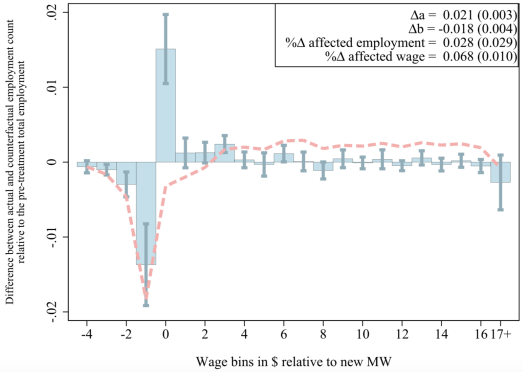
Cengiz et al. (2019)

Basic regression specification:

$$\frac{E_{sjt}}{N_{st}} = \sum_{\tau=-3}^{4} \sum_{k=-4}^{17} \alpha_{\tau k} I_{sjt}^{\tau k} + \mu_{sj} + \rho_{jt} + \Omega_{sjt} + u_{sjt}, \tag{2}$$

- $\frac{E_{sjt}}{N_{st}}$: ratio between employment in \$0.25 wage bins j in state s and at quarter t, and N_{st} is the size of the population in state s and quarter t.
- $I_{sjt}^{\tau k}$: which equals 1 if the minimum wage was raised τ years from date t and for the \$0.25 wage bins j that fall between k and $k+1$ dollars relative to the new minimum wage;
- μ_{sj}: state-by-wage-bin effects;
- ρ_{jt}: period-by-wage-bin effects;
- Ω_{sjt}: include controls for small or federal increases;
- u_{sjt} is the error term.

Cengiz et al. (2019)



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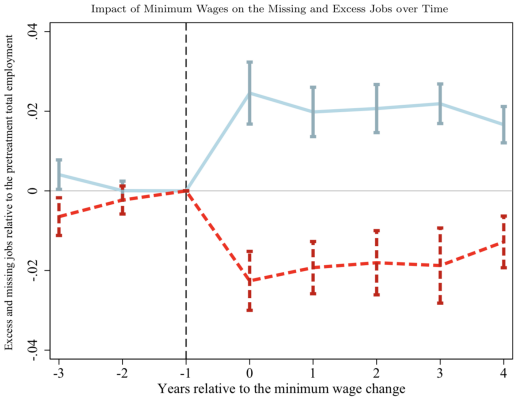

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Inference

The DiD setup (multiple regions with a single treatment date):

$$Y_t = \alpha + \delta_t + \sum_{r=1}^{\bar{R}} \gamma_r \mathbf{1}\{R = r\} + D_t\beta + V_t$$

where D_t varies at the region-time level.

The OLS estimator of β is the Best Linear Unbaised Estimator (i.e, efficient) if and only if there is no heteroskedasticity, no cross-section correlation and no serial correlation.

Issues with inference

1. **Clustering:** correlation within units i in the same group or region r (e.g. test scores are correlated within classes or schools since they are subject to common shocks);
2. **Serial correlation:** tendency for one observation to be correlated with past realizations (e.g.state-level employment rates are correlated over time)

Issues with inference: Clustering

Correlation within regions is often modeled using an additive random effects model:

$$V_{it} = \eta_{rt} + \varepsilon_{irt}$$

where $E(\eta_{rt}) = 0$ (equal trends) and $E(\varepsilon_{irt}) = 0$.

If we assume equal size of regions, the conventional OLS variance formula for β ignoring the clustered error structure:

$$\tilde{V}(\hat{\beta}) \propto \frac{\sigma_\varepsilon^2}{n},$$

where the correct sampling variance:

$$V(\hat{\beta}) \propto \sigma_\eta^2 + \frac{\sigma_\varepsilon^2}{n}.$$

Issues with inference: Clustering

By ignoring intra-class correlation, we overestimate the precision of the estimator by:

$$\frac{\tilde{V}(\hat{\beta})}{V(\hat{\beta})} = \frac{\sigma_\varepsilon^2}{n\sigma_\eta^2 + \sigma_\varepsilon^2}$$

- If $\sigma_\eta^2 = 0$ (there is no correlation within regions), then $\tilde{V}(\hat{\beta}) = V(\hat{\beta})$.
- the larger σ_η^2, the larger $V(\hat{\beta})$ relative to $\tilde{V}(\hat{\beta})$ (the individual level data is less informative).
- As a rule, the effective sample sizes are much closer to the number of regions than to the number of individual units in the dataset.

Issues with inference: Clustering

Here are some possible solutions for estimating the Var-Cov matrix:

1. Compute σ_η^2 and correct the conventional variance $\tilde{V}(\hat{\beta})$;
2. Cluster standard errors by region $\times$ time using the clustered variance estimator: $\hat{V}(\hat{\beta}) \propto \sigma_\eta^2 + \frac{\sigma_\varepsilon^2}{n}$;
3. Use region averages instead of micro data and estimate by weighted least squares using weights = $\frac{1}{\hat{V}(\hat{\beta})}$;
4. Use bootstrap inference by drawing blocks of data defined by region $\times$ time (block bootstrap)

Issues with inference: Serial Correlation

In most cases, the assumption that η_{rt} is serially uncorrelated is hard to defend (e.g. regional shocks tend to be positively auto-correlated). Hence, one should further allow for correlation in the η_{rt} themselves.

1. Allow for an unrestricted covariance structure over time within regions: i.e. cluster at the region-level instead of region $\times$ time-level (e.g. Cengiz et al., 2019)
2. With few clusters, use the Newey-West estimator that generalizes the heteroskedasticity-robust estimator of the variance for the presence of unknown auto-correlation patterns.

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De Chaisemartin and d'Haultfoeuille (2022)

De Chaisemartin, C., d'Haultfoeuille, X. (2022). Two-way fixed effects and differences-in-differences with heterogeneous treatment effects: A survey (No. w29691). National Bureau of Economic Research.
`https://www.nber.org/system/files/working_papers/w29691/w29691.pdf`

Abstract: Linear regressions with period and group fixed effects are widely used to estimate policies' effects: 26 of the 100 most cited papers published by the American Economic Review from 2015 to 2019 estimate such regressions. It has recently been show that those regressions may produce misleading estimates, if the policy's effect is heterogeneous between groups or over time, as is often the case. This survey reviews a fast-growing literature that documents this issue, and that proposes alternative estimators robust to heterogeneous effects.

de Chaisemartin and d'Haultfoeuille (2022)

Two-way fixed effect (TWFE) regressions is common in complicated designs with many groups and periods, variation in treatment timing, treatments switching on and off, and/or non-binary treatments.

In those more complicated designs, TWFE estimators are unbiased for an ATE:

1. if parallel trends holds, and
2. the treatment effect should be constant, between groups and over time.

de Chaisemartin and d'Haultfoeuille (2022)

In the usual TWFE regression:

$$Y_{i,g,t} = \hat{\alpha}_r + \hat{\gamma}_t + \hat{\beta}_{fe} D_{r,t} + \varepsilon_{r,t}$$

it can be shown that, under the parallel trend assumption on the potential outcome without treatment:

$$\mathbb{E}\left(\hat{\beta}_{fe}\right) = E\left[\sum_{(r,t):D_{r,t}\neq 0} W_{r,t} TE_{r,t}\right]$$

$TE_{r,t} = Y_{r,t}(1) - Y_{r,t}(0)$, the ATE in group r at time t.
from 0 to $D_{r,t}$, scaled by $D_{r,t}$.

de Chaisemartin and d'Haultfoeuille (2022)

What do the weights look like?

Take two regions (r_2, r_3) and three time periods (1: no treatment, 2: r_2 is treated, r_3 is not, 3: r_2, r_3 are treated). It can be shown that:

$$\hat{\beta}_{fe} = (DID_{r_2,r_3,1,2} + DID_{r_3,r_2,2,3})/2$$

with:

$$\begin{aligned} DID_{r_2,r_3,1,2} &= Y_{r_2,2} - Y_{r_2,1} - (Y_{r_3,2} - Y_{r_3,1}) \\ DID_{r_3,r_2,2,3} &= Y_{r_3,3} - Y_{r_3,r_3} - (Y_{r_2,3} - Y_{r_2,2}) \end{aligned}$$

Note that under the parallel trend assumption:

$$\mathbb{E}(DID_{r_2,r_3,1,2}) = \mathbb{E}(TE_{r_2,2})$$

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On the other hand:

$$\begin{aligned} Y_{r_2,3} - Y_{r_2,2} &= Y_{r_2,3}(0) + TE_{r_2,3} - (Y_{r_2,2}(0)) + TE_{r_2,2}) \\ Y_{r_3,3} - Y_{r_3,2} &= Y_{r_3,3}(0) + TE_{r_3,3} - Y_{r_3,2}(0) \end{aligned}$$

Hence, under the parallel trend assumption:

$$\mathbb{E}(DID_{r_3,r_2,2,3}) = \mathbb{E}(TE_{r_3,3} - TE_{r_2,3} + TE_{r_2,2})$$

and it follows that:

$$\mathbb{E}\left(\hat{\beta}_{fe}\right) = [1/2 TE_{r_3,3} + TE_{r_2,2} - 1/2 TE_{r_2,3}]$$

Comparison of 'treated in $t-1$ and t' to 'non-treated in $t-1$ and treated in t' are 'forbidden'.

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$$\mathbb{E}\left(\hat{\beta}_{fe}\right) = E\left[\sum_{(r,t):D_{r,t}\neq 0} W_{r,t} TE_{r,t}\right]$$

$W_{r,t}$ are weights summing to 1, that are proportional to and of the same sign as

$$N_{r,t}(D_{r,t} - D_{r,.} - D_{.,t} + D_{.,.}),$$

where:

- $N_{r,t}$ is the number of observations in cell (r,t),
- $D_{r,.}$ is the average treatment of group r across periods,
- $D_{.,t}$ is the average treatment at period t across groups,
- and $D_{.,.}$ is the average treatment across groups and periods

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Two implications:

1. $\hat{\beta}_{fe}$ may be biased for the ATT: the ATE across all treated (r, t) cells,
2. Some of the weights $W_{r,t}$ may be negative, and $\hat{\beta}_{fe}$ may be prone to "sign-reversal".

This means that in the minimum wage example, $\hat{\beta}_{fe}$ could be estimating something like:

$$3 \times \hat{\beta}_{\text{Santa Clara}} + (-2) \times \hat{\beta}_{\text{Wayne county}}$$

Then, if raising the minimum wage by one dollar leads to: $\hat{\beta}_{\text{Santa Clara}} = -5\% < 0$, and $\hat{\beta}_{\text{Wayne county}} = -20\% < 0$, one would have:

$$\mathbb{E}\left(\hat{\beta}_{fe}\right) = 3 \times (-0.05) + (-2) \times (-0.2) = 0.25 > 0$$

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Analyse the weights from a regression R-package twowayFEweights:
`https://cran.r-project.org/web/packages/TwoWayFEWeights/TwoWayFEWeights.pdf`

Special cases where $\hat{\beta}_{fe}$ is not biased:

1. (i) the design is staggered, meaning that groups' treatment can only increase over time and can change at most once; (ii) the treatment is binary; (iii) there is no variation in treatment timing: all treated groups start receiving the treatment at the same date.
2. $(D_{r,t} - D_{r,.} - D_{.,t} + D_{.,.})$ is uncorrelated with $TE_{r,t}$.

de Chaisemartin and d'Haultfoeuille (2022)

With a binary treatment de Chaisemartin and D'Haultfoeuille propose to use the DID_M estimator, a weighted average, across t, of two types of *DIDs*:

1. a DID comparing the $t-1$ to t outcome evolution of groups going from untreated to treated from $t-1$ to t, the "switchers in", and of groups untreated at both dates.
2. a DID comparing the $t-1$ to t outcome evolution of groups treated at both dates, and of groups going from treated to untreated from $t-1$ to t, the "switchers out".

See R-Package: DIDmultiplegt

Heterogeneity and multiple periods of treatment introduction

The main idea is to define treatment group- and time-specific ATTs:
the average effect in a specific outcome period (say $t+\tau$), among those subjects from whom the treatment is introduce at the beginning of a specific period t.

Let $G_t = 1$ if the region is treated in period $t_d = t$. This is grouping regions treated in the same period t. Define:

$$TE_{G_t=1,\tau} = \mathbb{E}\left[Y_{t+\tau}(1) - Y_{t+\tau}(0) | G_t = 1\right], \text{ with } \tau \geq 0, \text{ for } \tau \in \{0, 1, \ldots, T-1\}.$$

Heterogeneity and multiple periods of treatment introduction

To evaluate $TE_{G_t=1,\tau}$, the common trend assumption must hold when considering $G_t = 1$ as the treated group, and all subjects not treated up to $t_d = t$ as the control group.

$$E\left(Y_{t+\tau}(0) - Y_0(0) | G_t = 1\right) = E\left(Y_{t+\tau}(0) - Y_0(0) | G_{t^*} = 0, \forall t^* \leq t\right)$$

Under this condition, we may evaluate two types of aggregate effects:

$$ATT_{\text{group-specific}} = \frac{1}{T-t+1} \sum_{\tau=0}^{T-t} TE_{G_t=1,\tau}$$

$$ATT_{\text{time-specific}} = \sum_{t:t+\tau \leq T} \Pr\left(G_t = 1 | t+\tau \leq T\right) TE_{G_t=1,\tau}$$

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A word of caution

Be aware of the effects you are differencing out!

DiD models may be internally valid, because they control for a substantial part of the heterogeneity. However, they may wipe out a substantial part of the first order effects one is actually interested in.

One example of this is the discussion of Dustmann, Schönberg and Stuhler (2016) referring to a long controversy on the wage effect of migration.

Dustmann, Christian, Uta Schönberg, and Jan Stuhler. “The impact of immigration: Why do studies reach such different results?.” Journal of Economic Perspectives 30.4 (2016): 31-56.

Dustmann et al. (2016)

The existing empirical literature has derived three conceptually different effects of immigration on wages, estimated using different types of variation for identification:

1. estimation at the regional level exploiting variation in the total inflow of immigrants, as pioneered by Altonji and Card (1991);
2. estimation at the national level exploiting variation in the skill-cell-specific inflow of immigrants, as pioneered by Borjas (2003);
3. estimation at the regional level exploiting variation in the inflow of immigrants both across areas and skill cells, as for instance in Card (2001)

Dustmann et al. (2016)

The pure spatial approach: First difference expression:

$$\Delta \log w_{gart} = \theta_{ga}^{spatial} \Delta p_{rt} + (s_{ga} \times \Delta \pi_t) + \Delta \varphi_{gart}$$

- $\Delta \log w_{gart}$ change in native wage (in logs) in education group g, experience group a, region r and at time t,
- Δp_{rt} difference in the ratio of immigrants to all labour in each region r between two time periods
- s_{ga} and π_t are vectors of education-experience and time fixed effects.

In the case of two time periods and two regions A and B, the parameter $\theta_{ga}^{spatial}$ equals a difference-in-difference estimator where differences are taken over time and across regions.

Dustmann et al. (2016)

Provided that region B, otherwise identical to region A, did not experience an inflow of immigrants and is not indirectly affected by the immigration shock in region A (for example, through outmigration of natives), this parameter identifies the **total effect of immigration on wages** of a **particular skill group**.

It answers the question: “What is the overall effect of immigration on native wages of a particular education-experience group?”

Dustmann et al. (2016)

National skill-cell approach: First difference expression:

$$\Delta \log w_{gat} = \theta^{skill} \Delta p_{gat} + \Delta \pi_t + (s_g \times \Delta \pi_t) + (x_a \times \Delta \pi_t) + \Delta \varphi_{gat}$$

where:

- $\Delta \log w_{gat}$ change in native wage (in logs) in education group g, experience group a at time t,
- Δp_{gat} difference in the ratio of immigrants to all labour in each education-experience group ga between two time periods
- s_g , x_a , and π_t are vectors of education, experience, and time fixed effects.

θ^{skill} corresponds to a triple difference estimator where differences are taken over time, experience groups, and education groups.

Dustmann et al. (2016)

$\Delta\pi_t$ absorbs the overall immigration shock — any effects of immigration common to all education and experience groups are therefore differenced out.

The education-time fixed effects $(s_g \times \Delta\pi_t)$ capture, in addition to differential time trends by education unrelated to immigration, differences in immigration shocks across education groups.

Any effects of immigration **common to all experience groups within education groups** are therefore likewise differenced out.

The inclusion of experience-time fixed effects $(x_a \times \Delta\pi_t)$, in turn, soaks up the experience-specific immigration shock, in addition to allowing for differential time trends by experience unrelated to immigration.

Dustmann et al. (2016)

The parameter θ^{skill} therefore identifies the relative effect of immigration by experience and answers the question: “How does immigration affect native wages of experienced relative to inexperienced workers in the same education group?”

Since the effects of immigration that are common to the education group are differenced out, this parameter is **not informative** about the distributional effects between education groups, nor about its absolute effects.

Implicit in the interpretation: the main migration flows come from inexperienced low-skilled individuals.

Dustmann et al. (2016)

The mixture approach: relates the wage change of natives in education group g and region r to the education-specific immigration shock in that region (Δp_{grt}), controlling for education- and region-specific time trends $(s_g \times \Delta\pi_t)$ and $(s_r \times \Delta\pi_t)$.

$$\Delta \log w_{gart} = \theta^{spatial,skill} \Delta p_{grt} + (s_g \times \Delta\pi_t) + (s_r \times \Delta\pi_t) + \Delta\varphi_{grt}$$

In the simple case of two regions A and B, two time periods, and two education groups, the parameter $\theta^{spatial,skill}$ can be expressed as a triple difference estimator where differences are taken over time, across regions, and across education groups.

Dustmann et al. (2016)

The region-specific time effects absorbs the total region-specific immigration shock.

$\theta^{spatial,skill}$ identifies the relative effect of immigration by education and answers the question: "How does immigration affect native wages of low-skilled relative to high-skilled workers?"

Since the effects of immigration common to all education-experience groups are differenced out, the mixture approach is informative about the distributional effects of immigration between education groups, but not about its absolute effects.